Mikhail Trifonov

Java Backend Engineer

Innopolis, Russia +7 (961) 418-51-88 trifonov2812@gmail.com
github.com/LuminiteTime linkedin.com/in/mikhailtrifonov28



Profile

Backend engineer, 2.5 years of production Java/Scala: data platforms, microservices, distributed pipelines. Currently at T-Bank's Data Platform team, shipping a new Java/Kora data service end to end — data model, REST contracts, deployment, production support. Own components from design through release, and bring legacy code back under active development.

Technical Skills

Languages	Java, Scala, Python, SQL
Backend	Spring Boot, Spring Web, Spring Security, Spring Data, Hibernate, JDBC, Kora, Quarkus, FastAPI
Data & Storage	Apache Airflow, Apache Iceberg, Apache Arrow Flight, Kafka, DBqueue, PostgreSQL, Oracle, ClickHouse, MongoDB, Redis, S3, MinIO
Architecture	Microservices, event-driven systems, REST, gRPC, GraphQL, OpenAPI contracts
Infrastructure	Docker, Kubernetes, Terraform, Nginx, CI/CD, GitLab CI, GitHub Actions, Maven, Gradle
Observability	OpenTelemetry, Prometheus, Grafana, Loki
Testing	JUnit, Mockito, Testcontainers, testing pyramid, Postman
AI in workflow	Daily use of Codex and Claude for prototyping and research; build custom agent skills and editor workflows

Russian (native), English (B2).

Experience

T-Bank **Sep 2025 – Present**

Backend Engineer (Junior+), Data Platform

- Designing and shipping a new Java/Kora data service end to end: data model, REST contracts, deployment pipeline, and user support after release.
- Introduced end-to-end distributed tracing across all platform services using OpenTelemetry and Grafana, cutting incident investigation time.
- Redesigned and migrated a legacy GitOps file-processing pipeline, bringing unmaintained code back under active development.
- Built integrations with the GitLab API and two adjacent teams' APIs, including the pipeline streaming processing results into the company's Data Lake House (DLH); designed the user-facing data layout and access groups.
- Day-to-day stack: Java/Kora, PostgreSQL, Oracle, Kafka and DB queues for async flows, Apache Iceberg and Arrow Flight for large-scale storage and transfer, Python and Apache Airflow for orchestration.

Yandex, Auto.ru UGC Services **Jan 2025 – Apr 2025**

Backend Java/Scala Developer Intern

- Developed and maintained Scala microservices using gRPC for internal communication and REST APIs for external integrations.
- Reworked a custom gRPC client/server implementation to handle request hedging correctly on Scala, reducing response time by 75% on hedged calls.
- Designed and implemented GraphQL schemas for frontend-facing API interaction.

Hirus **Jul 2023 – Aug 2024**

Backend Java Developer

- Implemented Spring Web REST controllers and backend business logic for production medical software.
- Designed and optimized JDBC queries for PostgreSQL, reducing database load by 68%.
- Extended CI/CD: build pipelines, linters, and deployment steps.
- Covered service and controller layers with unit and integration tests (JUnit, Testcontainers), following the testing pyramid.

Hackathons

AdventureX 2026 — China's largest hackathon, Hangzhou, 800+ participants **Jul 2026**

- *The Sleepy Knight* — a pixel-art sleep tamagotchi whose hero stays alive only on the player's real sleep schedule. Built in 5 days with a 4-person team: React Native/Expo app in TypeScript on a Go backend.
- Owned the Go service behind the in-game AI oracle: structured-completion provider over a self-hosted OpenAI-compatible vLLM endpoint, input validation, retry on unparseable model output, request-scoped logging, and the GitHub Actions deploy

workflow.

- On the app side: game screens, onboarding and tutorial flows, Zustand state, event-driven local notifications, and an NFC e-ink companion card mirroring hero state outside the app; covered game state and reminder logic with Jest.

MTS True Tech Hack — Best Solution Award, MWS Tables track **Apr 2026**

- Built a Confluence-like collaborative document system with real-time editing, document storage, and native MWS Tables integration.
- Worked full-stack (React, TypeScript, Java/Spring); owned backend metadata architecture and the MWS Tables integration.
- Designed a PostgreSQL-based producer-consumer event flow between the Java metadata backend and the Node.js real-time editing service to keep them decoupled.
- Implemented authentication, document metadata, comments, versioning, user services, and OpenAPI contracts.

T1 Hackathon – PingTower — site & API uptime monitoring system **Sep 2025**

- Backend lead and system architect for a 4-day hackathon project: web dashboard, scheduled checks, configurable alert rules, incident history, and SLA reports.
- Owned the entire Java backend (Java 21, Spring Boot, PostgreSQL for relational data, ClickHouse for time-series analytics) and the OpenAPI contract that the frontend and notification service worked against.
- Split the system into two independent services behind a clean HTTP contract: Java backend for monitoring logic, Python service for notification delivery.

Techarena Kazan Challenge — 2nd Place **Jun 2025**

- Co-developed a C++ algorithm for distributing network flow between consumer nodes under capacity and cost constraints.
- Owned hypothesis testing and edge-case analysis; combined dynamic programming with a greedy strategy.

Selected Projects

Open Labs Share — peer-to-peer learning platform, backend architect **Jun 2025 – Jul 2025**

- Backend architect on a 7-person semester project; owned the user, auth, and gateway services.
- Built the user service from scratch (Spring Boot, Spring Data, Hibernate, PostgreSQL) behind Spring Cloud Gateway, with JWT authentication and Spring Security; designed gRPC and REST contracts between security and user services.
- Built a Marimo-style notebook runtime end to end: Java metadata service for state and ownership, plus a Python FastAPI sandbox for code execution.

Compilers Construction (Language I) — Java compiler for a custom imperative language **Spring 2025**

- Built a full compiler pipeline for a custom imperative language: JavaCC for lexing and parsing, hand-written semantic analysis passes, and code generation to WebAssembly Text Format (WAT).
- Implemented a CLI with `compile`, `run` (via wasmtime), and `ast` commands; covered the project with JUnit 5 unit tests plus a WebAssembly integration suite executing `.i` fixtures end-to-end.

Distributed Log Analysis Framework — MapReduce-based log processing **Apr 2025**

- Designed the architecture: Spring Boot API gateway, Go MapReduce analyzer, MongoDB storage, RabbitMQ task delivery, Prometheus and Grafana metrics; implemented the entire Java side.
- Built the Java gateway with Spring Data MongoDB, validation, OpenAPI docs; packaged everything in Docker Compose with synthetic log generators and load-tested the pipeline under several Python producers.

Education

Innopolis University **2023 – 2027**
Innopolis, Russia

B.Sc. in Computer Science, GPA: 4.77/5.00